THE DIMENSION OF HARMONIC MEASURE ON GALTON–WATSON TREES INCREASES AT SMALL BIAS

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Abstract. Let T be a Galton–Watson tree without leaves whose offspring number N is not constant and has finite mean m, and let d_λ be the Hausdorff dimension of the harmonic measure of the λ-biased random walk on T, $0 \le \lambda < m$. Lyons, Pemantle and Peres asked whether d_λ increases with λ. We prove that $\lambda \mapsto d_\lambda$ is continuously differentiable on $[0, k_0)$, where k_0 is the smallest value taken by N, and that the derivative of d_λ at $\lambda = 0$ equals $-\big(1 - \mathbb{E}[1/N]\big)\operatorname{Cov}\big(\log N, 1/N\big) > 0$. Hence d_λ is strictly increasing on an interval $[0, \varepsilon]$, with $\varepsilon > 0$ depending on the offspring law. No moment assumption beyond the finite mean is needed. The proof combines the formula of Lin and Rousselin for d_λ with a contraction estimate, uniform over trees, for the recursion satisfied by the escape probabilities and by their derivatives in λ.

1. Introduction

Let $(p_k)_{k\ge1}$ be a probability distribution on the positive integers with $p_k < 1$ for every k and finite mean $m = \sum_k k p_k$, so that $m > 1$, and let N denote a random variable with this distribution. Let T be a Galton–Watson tree with offspring distribution (p_k), rooted at e. Every vertex of T has at least one child. We write $\nu(v)$ for the number of children of a vertex v and

$$k_0 = \min\{k \ge 1 : \ p_k > 0\}$$

for the smallest possible number of children.

For $\lambda \ge 0$ the λ-biased random walk on T starts at the root, moves from the root to a uniformly chosen child, and moves from a vertex $v \ne e$ to its parent with probability $\lambda/(\lambda + \nu(v))$ and to each of its children with probability $1/(\lambda + \nu(v))$. Lyons [6] proved that the walk is transient exactly when $\lambda < m$. In that case the walk converges to a random ray of T; the law of this ray, given T, is the λ-harmonic measure on the boundary ∂T of T. The boundary carries the metric $\operatorname{dist}(\xi, \eta) = e^{-|\xi \wedge \eta|}$, where $|\xi \wedge \eta|$ is the number of edges common to the rays ξ and η. Lyons, Pemantle and Peres [7, 8] proved that the Hausdorff dimension of the λ-harmonic measure is almost surely equal to a constant d_λ, and that $d_\lambda < \log m$, the dimension of ∂T. For $\lambda = 0$ the walk never backtracks, the harmonic measure is the visibility measure of [7], and

$$d_0 = \mathbb{E}[\log N].$$

Lyons, Pemantle and Peres [9] asked whether d_λ is increasing in λ on $[0, m)$, and whether it is strictly increasing; this is Question 17.29 of [10]. Lin [5] and Rousselin [12] found, independently, an explicit stationary measure for the environment seen along the harmonic ray, and with it a formula for d_λ, recalled in Section 3. Lin deduced that $d_\lambda > d_0$ for every $\lambda \in (0, m)$, that $d_\lambda \to d_0$ as $\lambda \to 0$ and that $d_\lambda \to \log m$ as $\lambda \to m$. The comparison of d_λ and $d_{\lambda'}$ for

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$0 < \lambda < \lambda' < m$ is left open in [5], and the numerical computations of [12] support monotonicity. To our knowledge these are the only results on the dependence of d_λ on λ. Lin [5] also observes that a neighbouring quantity, the mean number of children of the vertices along the trajectory of the walk, is not monotone in λ.

The companion question for the speed of the walk, raised in [8, 9], is whether the speed on a Galton–Watson tree without leaves is nonincreasing in λ on $[0, m)$. It has been answered for small λ. Ben Arous, Fribergh and Sidoravicius [2] proved monotonicity when the bias away from the root is large enough. Aïdékon [1] gave an explicit formula for the speed, and with it the range was extended to $\lambda \le 1/2$, a bound due to Aïdékon and recorded in [13], and then to $\lambda \le k_0/\bigl(1+\sqrt{1-1/k_0}\bigr)$ when $k_0 \ge 2$ by Song, Wang and Xiang [13]. Mehrdad, Sen and Zhu [11] compare, for large bias, the speeds on two trees whose offspring distributions are stochastically ordered. The regularity of the speed in λ has been studied as well: Ben Arous, Hu, Olla and Zeitouni [3] proved, under an exponential moment assumption, the Einstein relation, which gives the derivative of the speed at the critical bias $\lambda = m$ in terms of the diffusivity of the critical walk, and Bowditch and Tokushige [4] proved that, on a supercritical Galton–Watson tree, the speed is differentiable at every λ for which the walk is ballistic and satisfies a central limit theorem. The monotonicity results above concern small λ, and the explicit ranges among them lie in $[0, k_0)$, which is also the range of Theorem 1 below.

Theorem 1. *The function $\lambda \mapsto d_\lambda$ is continuously differentiable on $[0, k_0)$, with the one-sided derivative at 0, and*

$$d_0' = \bigl(1 - \mathbb{E}[1/N]\bigr)\Bigl(\mathbb{E}[1/N]\,\mathbb{E}[\log N] - \mathbb{E}\bigl[N^{-1}\log N\bigr]\Bigr) = -\bigl(1-\mathbb{E}[1/N]\bigr)\operatorname{Cov}\bigl(\log N, 1/N\bigr). \tag{1}$$

This number is strictly positive.

Corollary 2. *There is $\varepsilon > 0$, depending on the offspring distribution, such that $d_\lambda < d_{\lambda'}$ whenever $0 \le \lambda < \lambda' \le \varepsilon$.*

Corollary 2 answers the question of [9], including strictness, on a neighbourhood of $\lambda = 0$. Since $1/N$ is bounded and $\mathbb{E}[\log N] \le m < \infty$, the right-hand side of (1) is finite under the standing assumptions; no second moment is used anywhere.

Example 3. If N takes the values 1 and 2 with probability $1/2$ each, then $\mathbb{E}[1/N] = 3/4$, $\mathbb{E}[\log N] = (\log 2)/2$ and $\mathbb{E}[N^{-1}\log N] = (\log 2)/4$, so $d_0' = (\log 2)/32$. If N takes the values 2 and 4 with probability $1/2$ each, then $d_0' = 5(\log 2)/128$. For the geometric distribution $p_k = 2^{-k}$ one has $\mathbb{E}[1/N] = \log 2$ and $d_0' = 0.05496$ to five decimal places.

Remark 4. In the formula of Lin and Rousselin, d_λ is the mean of $\log(\lambda + C_\lambda(T))$ under a probability measure on trees whose density with respect to the Galton–Watson measure is $\mathbb{E}[W_\lambda \mid T]/\mathbb{E}[W_\lambda]$, with W_λ defined in Section 3; here $C_\lambda(T)$ is the effective conductance of T. Both the function and the density depend on λ. At $\lambda = 0$ the derivative of $\log(\lambda + C_\lambda(T))$ is $\bigl(1 - \sum_{i \le \nu(e)} 1/\nu(e_i)\bigr)/\nu(e)$, where $e_1, \dots, e_{\nu(e)}$ are the children of the root. Its mean is zero, and it is negative with positive probability: this happens when the root has more than k_0 children and each of them has k_0 children. The slope (1) therefore comes entirely from the change of the density, which to first order favours trees whose root has many children; see (8).

The escape probability of the walk from a tree is the fixed point of a recursion over the subtrees of the children of the root. Section 2 shows that, written for the logarithm of the escape probability and restricted to trees in which every vertex has at least k children, this recursion is a contraction in the supremum norm with constant λ/k, uniformly in the tree, and that the recursion obtained by differentiating in λ has the same contraction coefficient in its

derivative variables. This gives continuous differentiability of the escape probability of every such tree on $[0,k)$, with a bound on the derivative that does not depend on the tree, and allows differentiation of the formula for d_λ under the expectation. Section 3 contains the proof of Theorem 1, and Section 4 reports a numerical evaluation of d_λ for four offspring distributions.

2. Escape probabilities

For an integer $k \geq 1$ let $\mathcal{T}_k$ be the set of rooted trees in which every vertex has finitely many and at least k children. For $T \in \mathcal{T}_1$ let T_* be the tree obtained by joining a new vertex e_* to the root e of T, so that e_* is the parent of e. For $\lambda \geq 0$ consider the walk on T_* that starts at e and moves from every vertex $v \neq e_*$ to its parent with probability $\lambda/(\lambda+\nu(v))$ and to each child with probability $1/(\lambda+\nu(v))$. For $j \geq 0$ let

$$b_j(T,\lambda) = \mathbb{P}\big(\text{the walk reaches generation } j \text{ of } T \text{ before } e_*\big),$$
$$\beta_\lambda(T) = \mathbb{P}\big(\text{the walk never visits } e_*\big),$$

so that $b_0 = 1$. If the root of T has n children, we write $T_1,\dots,T_n$ for the subtrees of T rooted at these children; they belong to $\mathcal{T}_k$ when T does.

Lemma 5. *Let $T \in \mathcal{T}_1$, $\lambda \geq 0$ and $j \geq 1$, and let $S = \sum_{i=1}^n b_{j-1}(T_i,\lambda)$. Then*

$$b_j(T,\lambda) = \frac{S}{\lambda+S}, \tag{2}$$

and $b_j(T,\lambda)$ decreases to $\beta_\lambda(T)$ as $j \to \infty$.

Proof. The first j generations of T form a finite set. For $\lambda > 0$ the walk leaves this set almost surely, because from each of its vertices the walk reaches e_* in at most j steps with probability bounded below; for $\lambda = 0$ the walk moves away from the root at every step. Hence, almost surely, the walk reaches e_* or generation j. Condition on the first step, which leads to e_* with probability $\lambda/(\lambda+n)$ and to the ith child with probability $1/(\lambda+n)$. From the ith child the walk reaches generation j of T before returning to e with probability $b_{j-1}(T_i,\lambda)$, and otherwise it returns to e and starts afresh. Therefore $b = b_j(T,\lambda)$ satisfies

$$b = \frac{1}{\lambda+n}\sum_{i=1}^n \Big(b_{j-1}(T_i,\lambda) + \big(1 - b_{j-1}(T_i,\lambda)\big)b \Big),$$

that is $b(\lambda+S) = S$. Here $\lambda + S > 0$: this is clear if $\lambda > 0$, and if $\lambda = 0$ then every b_{j-1} equals 1 and $S = n$. So (2) holds. The events in the definition of b_j decrease with j, and by the first sentence of the proof their intersection is, up to a null event, the event that the walk never visits e_*. □

Lemma 6. *Let $k \geq 1$, $T \in \mathcal{T}_k$ and $0 \leq \lambda \leq k$. Then $b_j(T,\lambda) \geq 1 - \lambda/k$ for every $j \geq 0$, hence $\beta_\lambda(T) \geq 1 - \lambda/k$. If $\lambda < k$, then*

$$\beta_\lambda(T) = \frac{C_\lambda(T)}{\lambda + C_\lambda(T)}, \qquad C_\lambda(T) = \sum_{i=1}^n \beta_\lambda(T_i). \tag{3}$$

Proof. The bound holds for $j = 0$. If it holds for $j-1$ and all trees of $\mathcal{T}_k$, then in (2) we have $S \geq n(1-\lambda/k) \geq k - \lambda$, and since $s \mapsto s/(\lambda+s)$ is nondecreasing, $b_j(T,\lambda) \geq (k-\lambda)/k$. Letting $j \to \infty$ in (2), which involves finitely many subtrees and a denominator bounded below by $k - \lambda > 0$, gives (3). □

For $\lambda > 0$ the quantity $C_\lambda(T)$ is the effective conductance between the root of T and infinity when an edge between generations j and $j+1$ has conductance λ^{-j}.

Proposition 7. *Let $k \geq 1$ and $0 < r < k$, and put $c = 1 - r/k$ and $M = k/(k-r)^2$. For every $T \in \mathcal{T}_k$ the function $\lambda \mapsto \beta_\lambda(T)$ is continuously differentiable on $[0, r]$, and for all $\lambda \in [0, r]$,*

$$c \leq \beta_\lambda(T) \leq 1, \qquad \big|\partial_\lambda \beta_\lambda(T)\big| \leq M, \qquad \partial_\lambda \beta_\lambda(T)\big|_{\lambda=0} = -\frac{1}{\nu(e)}.$$

Proof. Throughout, T ranges over $\mathcal{T}_k$ and λ over $[0, r]$, and $\|\cdot\|$ denotes the supremum over all such pairs. Put $\rho = r/k < 1$ and $x_j = \log b_j$. By Lemma 6, $\log c \leq x_j \leq 0$.

The recursion in logarithmic coordinates. For $n \geq k$ and $x \in [\log c, 0]^n$ let $S(x) = \sum_i e^{x_i}$ and $H_\lambda(x) = \log S(x) - \log(\lambda + S(x))$, so that by (2)

$$x_j(T, \lambda) = H_\lambda\big(x_{j-1}(T_1, \lambda), \dots, x_{j-1}(T_n, \lambda)\big).$$

We have

$$\frac{\partial H_\lambda}{\partial x_i} = q\, w_i, \qquad q = \frac{\lambda}{\lambda + S}, \quad w_i = \frac{e^{x_i}}{S}, \qquad \frac{\partial H_\lambda}{\partial \lambda} = -\frac{1}{\lambda + S}. \tag{4}$$

The w_i are nonnegative and sum to 1. Since $S \geq nc \geq k - r$, we have $q \leq r/(r + k - r) = \rho$ and $1/(\lambda + S) \leq 1/(k - r)$. The cube $[\log c, 0]^n$ is convex, so H_λ is ρ-Lipschitz on it for the maximum norm. Consequently $\delta_j = \|x_j - x_{j-1}\|$ satisfies $\delta_j \leq \rho\, \delta_{j-1}$ for $j \geq 2$, and $\delta_1 = \sup \log((\lambda + n)/n) \leq \log(1 + \rho) \leq \rho$. Thus $\delta_j \leq \rho^j$.

The derivatives. For fixed T and j, the function $\lambda \mapsto b_j(T, \lambda)$ depends on the first j generations of T only and, by (2) and induction, is a rational function of λ that is positive on $[0, r]$. Let $y_j = \partial_\lambda x_j$, so that $y_0 = 0$. By the chain rule and (4),

$$y_j(T, \lambda) = -\frac{1}{\lambda + S_j} + q_j \sum_{i=1}^{n} w_{j,i}\, y_{j-1}(T_i, \lambda), \tag{5}$$

where S_j, q_j and $w_{j,i}$ are computed from the vector $(x_{j-1}(T_i, \lambda))_{i \leq n}$. Hence $\|y_j\| \leq 1/(k-r) + \rho\|y_{j-1}\|$, and by induction

$$\|y_j\| \leq \frac{1}{(k - r)(1 - \rho)} = M \qquad \text{for all } j.$$

For $j \geq 2$ subtract (5) for $j - 1$ from (5) for j:

$$y_j - y_{j-1} = \Big(\frac{1}{\lambda + S_{j-1}} - \frac{1}{\lambda + S_j}\Big) + q_j \sum_i w_{j,i}\big(y_{j-1} - y_{j-2}\big)(T_i) + \sum_i \big(q_j w_{j,i} - q_{j-1} w_{j-1,i}\big) y_{j-2}(T_i).$$

Since $|e^s - e^t| \leq |s - t|$ for $s, t \leq 0$, we have $|b_{j-1}(T_i) - b_{j-2}(T_i)| \leq \delta_{j-1}$ for every i and $|S_j - S_{j-1}| \leq n\delta_{j-1}$. All of S_j, S_{j-1} are at least nc. The first bracket is therefore at most $n\delta_{j-1}/(nc)^2 \leq \delta_{j-1}/(kc^2)$ in absolute value. The second term is at most $\rho\|y_{j-1} - y_{j-2}\|$. For the third, write $q\, w_i = \lambda b_i/(S(\lambda + S))$ with $b_i = e^{x_i}$, and let $\tilde{b}_i$, $\tilde{S}$ be the corresponding quantities one step earlier. Then

$$\begin{aligned}\sum_i \Big|\frac{b_i}{S(\lambda + S)} - \frac{\tilde{b}_i}{\tilde{S}(\lambda + \tilde{S})}\Big| &\leq \frac{\sum_i |b_i - \tilde{b}_i|}{S(\lambda + S)} + \tilde{S}\,\Big|\frac{1}{S(\lambda + S)} - \frac{1}{\tilde{S}(\lambda + \tilde{S})}\Big| \\ &= \frac{\sum_i |b_i - \tilde{b}_i|}{S(\lambda + S)} + \frac{|S - \tilde{S}|\,(\lambda + S + \tilde{S})}{S(\lambda + S)(\lambda + \tilde{S})} \leq \frac{n\delta_{j-1}}{(nc)^2} + \frac{2n\delta_{j-1}}{(nc)^2},\end{aligned}$$

where the last step uses $\lambda + S + \tilde{S} \leq (\lambda + S) + (\lambda + \tilde{S})$. Multiplying by $\lambda \leq r$, the third term is at most $3r\delta_{j-1}M/(kc^2)$. Altogether, with $K = (1 + 3rM)/(kc^2)$ and $\theta_j = \|y_j - y_{j-1}\|$,

$$\theta_j \leq \rho\, \theta_{j-1} + K\delta_{j-1} \leq \rho\, \theta_{j-1} + K\rho^{j-1} \qquad (j \geq 2),$$

and $\theta_1 = \|y_1\| \leq 1/(k - r)$. By induction $\theta_j \leq \rho^{j-1}\big(\theta_1 + (j-1)K\big)$, so $\sum_j \theta_j < \infty$.

Conclusion. The functions y_j converge uniformly on $\mathcal{T}_k \times [0,r]$ to a function y with $\|y\| \le M$. For fixed T each $y_j(T,\cdot)$ is continuous on $[0,r]$, hence so is $y(T,\cdot)$. By Lemma 5, $x_j(T,\lambda) \to x(T,\lambda) = \log \beta_\lambda(T)$. Passing to the limit in $x_j(T,\lambda) = x_j(T,0) + \int_0^\lambda y_j(T,t)\,dt$, and noting that $x_j(T,0) = 0$, we get $x(T,\lambda) = \int_0^\lambda y(T,t)\,dt$. Thus $\lambda \mapsto \log \beta_\lambda(T)$ is continuously differentiable on $[0,r]$ with derivative $y(T,\cdot)$, and $|\partial_\lambda \beta_\lambda| = \beta_\lambda |y| \le M$. Finally, at $\lambda = 0$ we have $b_{j-1} \equiv 1$, $S_j = n$ and $q_j = 0$ in (5), so $y_j(T,0) = -1/n$ for every $j \ge 1$, and the same holds for $y(T,0) = \partial_\lambda \beta_\lambda(T)|_{\lambda=0}$. □

3. Proof of Theorem 1

Let T and $\tilde{T}$ be independent Galton–Watson trees with offspring distribution (p_k). Almost surely both belong to $\mathcal{T}_{k_0}$, as do all the subtrees rooted at their vertices. For $0 \le \lambda < k_0$ put

$$B_\lambda = \beta_\lambda(\tilde{T}), \qquad C_\lambda = C_\lambda(T), \qquad W_\lambda = \frac{B_\lambda C_\lambda}{\lambda - 1 + B_\lambda + C_\lambda}, \qquad L_\lambda = \log(\lambda + C_\lambda).$$

For $0 < \lambda < m$ the same definitions make sense, with β_λ and C_λ related as in (3), and Lin [5, formulas (7) and (8) and the display that follows (9)] proved that

$$d_\lambda = \frac{\mathbb{E}[W_\lambda L_\lambda]}{\mathbb{E}[W_\lambda]} \qquad (0 < \lambda < m); \tag{6}$$

see also [12, Theorem 4.1]. The function $T \mapsto \mathbb{E}[W_\lambda \mid T]/\mathbb{E}[W_\lambda]$ is the density, with respect to the Galton–Watson measure, of a probability measure on trees that is stationary for the environment seen along the harmonic ray. At $\lambda = 0$ we have $\beta_0 \equiv 1$, hence $C_0 = \nu(e)$, $W_0 = 1$ and $L_0 = \log \nu(e)$, and the right-hand side of (6) is $\mathbb{E}[\log N] = d_0$. So (6) holds for $\lambda = 0$ as well.

Proof of Theorem 1. Fix $r \in (0, k_0)$ and let $c = 1 - r/k_0$ and $M = k_0/(k_0 - r)^2$. Write $n = \nu(e)$ for the number of children of the root of T and $D_\lambda = \lambda - 1 + B_\lambda + C_\lambda$. Primes denote derivatives in λ. By Proposition 7, applied to $\tilde{T}$ and to $T_1, \dots, T_n$, the functions B_λ and C_λ are continuously differentiable on $[0,r]$ and

$$c \le B_\lambda \le 1, \qquad cn \le C_\lambda \le n, \qquad |B'_\lambda| \le M, \qquad |C'_\lambda| \le nM.$$

By Lemma 6, $B_\lambda \ge 1 - \lambda/k_0$ and $C_\lambda \ge n(1 - \lambda/k_0)$, so

$$D_\lambda \ge \lambda\Big(1 - \frac{1}{k_0}\Big) + n\Big(1 - \frac{\lambda}{k_0}\Big) \ge cn, \qquad D_\lambda \le r + n.$$

Therefore W_λ and L_λ are continuously differentiable on $[0,r]$, with

$$W'_\lambda = \frac{B'_\lambda C_\lambda + B_\lambda C'_\lambda}{D_\lambda} - \frac{B_\lambda C_\lambda (1 + B'_\lambda + C'_\lambda)}{D_\lambda^2}, \qquad L'_\lambda = \frac{1 + C'_\lambda}{\lambda + C_\lambda},$$

and, using $n \ge 1$,

$$\frac{c^2}{1+r} \le W_\lambda \le \frac{1}{c}, \qquad |W'_\lambda| \le \frac{2M}{c} + \frac{1+2M}{c^2},$$

$$|L_\lambda| \le \log n + \kappa, \quad \kappa = \max\big(\log(1+r), \log(1/c)\big), \qquad |L'_\lambda| \le \frac{1+M}{c}.$$

Hence $|(W_\lambda L_\lambda)'|$ and $|W'_\lambda|$ are bounded on $[0,r]$ by a constant multiple of $1 + \log \nu(e)$, which is integrable because $\mathbb{E}[\log N] \le m$. By the mean value theorem and dominated convergence, $\lambda \mapsto \mathbb{E}[W_\lambda L_\lambda]$ and $\lambda \mapsto \mathbb{E}[W_\lambda]$ are continuously differentiable on $[0,r]$, with derivatives $\mathbb{E}[(W_\lambda L_\lambda)']$

and $\mathbb{E}[W'_\lambda]$. Since $\mathbb{E}[W_\lambda] \geq c^2/(1+r)$, the quotient (6) is continuously differentiable on $[0,r]$. As $r < k_0$ was arbitrary, this proves the first assertion, and

$$d'_\lambda = \frac{\mathbb{E}[W'_\lambda L_\lambda] + \mathbb{E}[W_\lambda L'_\lambda] - d_\lambda\,\mathbb{E}[W'_\lambda]}{\mathbb{E}[W_\lambda]} \qquad (0 \leq \lambda < k_0). \tag{7}$$

It remains to evaluate (7) at $\lambda = 0$. Let $N = \nu(e)$ be the number of children of the root of T, let $N_1, \dots, N_N$ be the numbers of children of these children, let $\tilde{N}$ be the number of children of the root of $\tilde{T}$, and put $A = \sum_{i \leq N} 1/N_i$. By Proposition 7,

$$B_0 = 1, \quad B'_0 = -\frac{1}{\tilde{N}}, \qquad C_0 = N, \quad C'_0 = -A, \qquad D_0 = N, \quad D'_0 = 1 - \frac{1}{\tilde{N}} - A.$$

Substituting,

$$W'_0 = \frac{-N/\tilde{N} - A}{N} - \frac{N(1 - 1/\tilde{N} - A)}{N^2} = -\frac{1}{\tilde{N}} - \Big(1 - \frac{1}{\tilde{N}}\Big)\frac{1}{N}, \qquad L'_0 = \frac{1-A}{N}; \tag{8}$$

the terms containing A cancel in W'_0. Let $a = \mathbb{E}[1/N]$ and $\ell = \mathbb{E}[\log N]$. The variable $\tilde{N}$ is independent of N, and given N the N_i are independent copies of N, so $\mathbb{E}[A \mid N] = Na$. Hence

$$\mathbb{E}[L'_0] = a - a = 0, \qquad \mathbb{E}[W'_0] = -2a + a^2, \qquad \mathbb{E}[W'_0 \log N] = -a\ell - (1-a)\mathbb{E}\big[N^{-1}\log N\big].$$

With $W_0 = 1$, $L_0 = \log N$ and $d_0 = \ell$, formula (7) gives

$$d'_0 = -a\ell - (1-a)\mathbb{E}\big[N^{-1}\log N\big] + (2a - a^2)\ell = (1-a)\Big(a\ell - \mathbb{E}\big[N^{-1}\log N\big]\Big),$$

which is (1).

For the sign, let N and $\tilde{N}$ be independent with law (p_k). Then

$$-2\,\mathrm{Cov}\big(\log N, 1/N\big) = \mathbb{E}\Big[\big(\log N - \log \tilde{N}\big)\Big(\frac{1}{\tilde{N}} - \frac{1}{N}\Big)\Big].$$

The integrand is nonnegative, since the logarithm is increasing and $s \mapsto 1/s$ is decreasing, and it is strictly positive on the event $\{N \neq \tilde{N}\}$, which has positive probability because N is not constant. Finally $a < 1$ because $\mathbb{P}(N \geq 2) > 0$. Hence $d'_0 > 0$. □

Proof of Corollary 2. By Theorem 1 the derivative of d_λ is continuous on $[0, k_0)$ and positive at 0, so it is positive on some interval $[0, \varepsilon]$. □

Remark 8. Let Q_λ be the probability measure with density $W_\lambda/\mathbb{E}[W_\lambda]$ with respect to the law of $(T, \tilde{T})$. Formula (7) reads

$$d'_\lambda = \mathbb{E}_{Q_\lambda}\Big[\frac{1 + C'_\lambda}{\lambda + C_\lambda}\Big] + \mathrm{Cov}_{Q_\lambda}\Big(\log(\lambda + C_\lambda), \frac{W'_\lambda}{W_\lambda}\Big) \qquad (0 \leq \lambda < k_0).$$

The derivatives B'_λ and C'_λ are nonpositive, since the escape probabilities decrease in λ, but this does not determine the sign of either term. At $\lambda = 0$ the first term vanishes and the second equals (1).

4. Numerical evaluation

Formula (6) can be evaluated by simulation of the recursion (3). A pool of 2×10^5 pairs, representing samples of $(\beta_\lambda(T), \partial_\lambda \beta_\lambda(T))$, is initialised at $(1, 0)$ and updated 300 times: every new pair is computed from (3) and its derivative in λ, with an independent offspring number and children drawn independently and uniformly from the current pool. For the evaluation, 2×10^5 roots are drawn, each with a fresh offspring number and children taken from the final pool, which give C_λ and C'_λ, and with one further independent element of the pool, which gives B_λ and B'_λ. The expectations in (6) and (7) are replaced by averages over these roots; only

the final pool is used. Each run was repeated with the six seeds $1000, \dots, 1005$, and for a given seed the same random numbers were used for every λ, so that the runs at different values of λ are coupled. Every $\pm$ below is the sample standard deviation of the six seed results divided by $\sqrt{6}$. The program and its output are in the repository listed at the end of the paper.

The bias runs over the grid $\lambda/m \in \{0, 0.02, 0.05, 0.1, 0.2, \dots, 0.9\}$, for the four offspring distributions of Tables 1 and 2.

offspring law	m	d_0	d_0'	pool average	quotient
$p_1 = p_2 = \frac{1}{2}$	1.500	0.34657	0.02166	0.02206 ± 0.00041	0.02228 ± 0.00041
$p_2 = p_4 = \frac{1}{2}$	3.000	1.03972	0.02708	0.02720 ± 0.00019	0.02718 ± 0.00019
$p_1 = p_2 = p_3 = \frac{1}{3}$	2.000	0.59725	0.04954	0.04925 ± 0.00020	0.04954 ± 0.00019
$p_k = 2^{-k}$	2.000	0.50783	0.05496	0.05572 ± 0.00062	0.05677 ± 0.00061

TABLE 1. The slope at $\lambda = 0$. The fourth column is (1), the fifth is the pool average of the right-hand side of (7) at $\lambda = 0$, and the last is $(d_\lambda - d_0)/\lambda$ at $\lambda = 0.02\, m$, with both terms pool estimates for the same seed; the last two are means over six seeds with their standard errors.

λ/m	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$p_1 = p_2 = \frac{1}{2}$	0.3466	0.3501	0.3540	0.3587	0.3643	0.3709	0.3787	0.3872	0.3954	0.4021
$p_2 = p_4 = \frac{1}{2}$	1.0394	1.0475	1.0556	1.0636	1.0713	1.0785	1.0850	1.0905	1.0948	1.0977
$p_1 = p_2 = p_3 = \frac{1}{3}$	0.5978	0.6080	0.6190	0.6310	0.6436	0.6561	0.6676	0.6775	0.6854	0.6909
$p_k = 2^{-k}$	0.5080	0.5203	0.5356	0.5548	0.5786	0.6056	0.6324	0.6559	0.6747	0.6878

TABLE 2. Estimates of d_λ, means over six seeds. The column $\lambda = 0$ is the pool estimate, to be compared with the exact d_0 of Table 1. The standard errors, over the whole grid, are at most 0.00050.

In Table 1 the pool averages agree with (1) to within one and a half standard errors, and the difference quotients, taken at the positive bias $0.02\, m$, are within 0.002 of it. A difference quotient at positive bias exceeds d_0' by about $0.01\, m\, d_0''$; for the geometric law d_0'' is about 0.052, which accounts for 0.0010 of the difference 0.0018 there. In Table 2 the estimates increase along the grid and stay below $\log m$, which equals 0.4055, 1.0986, 0.6931 and 0.6931 for the four distributions. The increase holds for every one of the six seeds and every pair of consecutive grid points. This agrees with the computations of [12] and with a positive answer to the question of [9] on the whole interval $[0, m)$, which Theorem 1 leaves open for λ away from 0.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [14] and working together inside the Brain framework [16], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [15], the Math Brain repository [14] and the Brain framework repository [16].

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [16] and Math Brain [14]. In October 2026 AI agents working from the project's list of open problems in probability obtained Theorem 1 and proofs of it, and a further agent checked the arguments. In November 2026 a Claude Fable agent (Anthropic) checked the proof again, wrote the simulation program and prepared the manuscript. GPT-6.1 in Codex (OpenAI) then checked the proofs and recomputed the numerical tables independently. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Programs and outputs:
`https://github.com/yspennstate/harmonic-dimension-small-bias`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [14].

Appendix A. Notation

Symbols are listed in the order in which they first appear. Primes denote derivatives in λ.

Symbol	Meaning	First appears
(p_k), m	the offspring distribution and its mean $m = \sum_k k p_k$	Section 1
N	a random variable with law (p_k); in Section 3, the number of children of the root of T	Section 1
T, e	a Galton–Watson tree with offspring distribution (p_k), and its root	Section 1
$\nu(v)$	the number of children of a vertex v	Section 1
k_0	the smallest possible number of children, $\min\{k \geq 1 : p_k > 0\}$	Section 1
λ	the bias: the walk moves from v to its parent with probability $\lambda/(\lambda + \nu(v))$	Section 1
∂T, ξ, η	the boundary of T and two rays of T	Section 1
dist	the metric $e^{-\lvert\xi\wedge\eta\rvert}$ on ∂T	Section 1
d_λ	the Hausdorff dimension of the λ-harmonic measure	Section 1
$\mathbb{E}$, $\mathbb{P}$, Cov	expectation, probability and covariance	Section 1
d'_λ, d'_0, d''_0	derivatives of d_λ in λ, and their values at $\lambda = 0$	Theorem 1
ε	the length of the interval in Corollary 2	Corollary 2

Symbol	Meaning	First appears
$C_\lambda(T)$	$\sum_{i\le n}\beta_\lambda(T_i)$, the effective conductance of T, defined in (3)	Remark 4
W_λ	the weight in the formula of Lin and Rousselin, defined in Section 3	Remark 4
$\mathcal{T}_k$, k	the rooted trees in which every vertex has finitely many and at least k children	Section 2
T_*, e_*	T with a new vertex e_* joined to its root as the root's parent	Section 2
$b_j(T,\lambda)$	the probability that the walk on T_* from e reaches generation j of T before e_*	Section 2
$\beta_\lambda(T)$	the probability that the walk on T_* from e never visits e_*	Section 2
n, $T_1,\dots,T_n$	the number of children of the root, and the subtrees rooted at them	Section 2
S	$\sum_{i\le n} b_{j-1}(T_i,\lambda)$; in the proof of Proposition 7, $S(x)=\sum_i e^{x_i}$	Lemma 5
r, c, M	$0<r<k$, $c=1-r/k$, $M=k/(k-r)^2$	Proposition 7
$\|\cdot\|$	the supremum over $T\in\mathcal{T}_k$ and $\lambda\in[0,r]$	proof of Proposition 7
ρ	r/k	proof of Proposition 7
x_j, x	$\log b_j$; a point of the cube $[\log c,0]^n$	proof of Proposition 7
H_λ	$\log S(x)-\log(\lambda+S(x))$, the recursion in logarithmic coordinates	proof of Proposition 7
q, w_i	$\lambda/(\lambda+S)$ and e^{x_i}/S	(4)
δ_j	$\|x_j-x_{j-1}\|$	proof of Proposition 7
y_j, y	$\partial_\lambda x_j$ and its limit	proof of Proposition 7
S_j, q_j, $w_{j,i}$	S, q, w_i computed from $(x_{j-1}(T_i,\lambda))_{i\le n}$	(5)
b_i, $\tilde b_i$, $\tilde S$	e^{x_i}, and b_i and S one step earlier	proof of Proposition 7
K, θ_j	$(1+3rM)/(kc^2)$ and $\|y_j-y_{j-1}\|$	proof of Proposition 7
$\tilde T$	a Galton–Watson tree independent of T	Section 3
B_λ, L_λ, D_λ	$\beta_\lambda(\tilde T)$, $\log(\lambda+C_\lambda)$, $\lambda-1+B_\lambda+C_\lambda$	Section 3
κ	$\max(\log(1+r),\log(1/c))$	proof of Theorem 1
N_i, $\tilde N$, A	the numbers of children of the children of the root of T, of the root of $\tilde T$, and $\sum_{i\le N}1/N_i$	proof of Theorem 1
a, ℓ	$\mathbb{E}[1/N]$ and $\mathbb{E}[\log N]$	proof of Theorem 1
Q_λ	the probability measure with density $W_\lambda/\mathbb{E}[W_\lambda]$ with respect to the law of $(T,\tilde T)$	Remark 8

References

[1] E. Aïdékon, Speed of the biased random walk on a Galton–Watson tree, *Probab. Theory Related Fields* **159** (2014), 597–617, doi:10.1007/s00440-013-0515-y.

[2] G. Ben Arous, A. Fribergh and V. Sidoravicius, Lyons–Pemantle–Peres monotonicity problem for high biases, *Comm. Pure Appl. Math.* **67** (2014), 519–530, doi:10.1002/cpa.21505.

[3] G. Ben Arous, Y. Hu, S. Olla and O. Zeitouni, Einstein relation for biased random walk on Galton–Watson trees, *Ann. Inst. Henri Poincaré Probab. Stat.* **49** (2013), 698–721, doi:10.1214/12-AIHP486.

[4] A. Bowditch and Y. Tokushige, Differentiability of the speed of biased random walks on Galton–Watson trees, *ALEA Lat. Am. J. Probab. Math. Stat.* **17** (2020), 609–642, doi:10.30757/ALEA.v17-24.

[5] S. Lin, Harmonic measure for biased random walk in a supercritical Galton–Watson tree, *Bernoulli* **25** (2019), 3652–3672, doi:10.3150/19-BEJ1106.

[6] R. Lyons, Random walks and percolation on trees, *Ann. Probab.* **18** (1990), 931–958, doi:10.1214/aop/1176990730.

[7] R. Lyons, R. Pemantle and Y. Peres, Ergodic theory on Galton–Watson trees: speed of random walk and dimension of harmonic measure, *Ergodic Theory Dynam. Systems* **15** (1995), 593–619, doi:10.1017/S0143385700008543.

[8] R. Lyons, R. Pemantle and Y. Peres, Biased random walks on Galton–Watson trees, *Probab. Theory Related Fields* **106** (1996), 249–264, doi:10.1007/s004400050064.

[9] R. Lyons, R. Pemantle and Y. Peres, Unsolved problems concerning random walks on trees, in *Classical and Modern Branching Processes*, IMA Vol. Math. Appl. **84**, Springer, 1997, 223–237, doi:10.1007/978-1-4612-1862-3_18.

[10] R. Lyons and Y. Peres, *Probability on Trees and Networks*, Cambridge University Press, 2016, doi:10.1017/9781316672815. Question numbers are those of the version of 19 August 2026 on the first author's web page.

[11] B. Mehrdad, S. Sen and L. Zhu, The speed of a biased walk on a Galton–Watson tree without leaves is monotonic with respect to progeny distributions for high values of bias, *Ann. Inst. Henri Poincaré Probab. Stat.* **51** (2015), 304–318, doi:10.1214/13-AIHP573.

[12] P. Rousselin, Invariant measures, Hausdorff dimension and dimension drop of some harmonic measures on Galton–Watson trees, *Electron. J. Probab.* **23** (2018), 1–31, doi:10.1214/18-EJP170.

[13] H. Song, L. Wang and K. Xiang, The speed of a biased walk on a Galton–Watson tree without leaves is monotonic for low values of bias, *J. Appl. Probab.* **62** (2025), 1044–1052, doi:10.1017/jpr.2024.113.

[14] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, `https://github.com/yspennstate/math-brain-public`.

[15] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, `https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24`, archived with doi:10.5281/zenodo.24150837.

[16] Y. Shmalo, *Brain Framework*, public software repository, 2026, `https://github.com/yspennstate/brain-framework-public`.

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